PORTFOLIO / SOUND DESIGN 2026

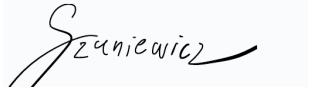
APPLICATION PORTFOLIO

IGOR SZUNIEWICZ

SOUND DESIGNER / AUDIO PROGRAMMER

FH JOANNEUM & Kunstuniversität Graz
Institute of Design & Communication
Master's Programme in Sound Design
Graz, Austria

→ **igorszuniewicz.com**



Kortrijk, 2026

Igor Szuniewicz

SOUND DESIGNER /
AUDIO PROGRAMMER

Contact

igorszuniewiczwork@gmail.com

igorszuniewicz.com

+48 605 117 516

linkedin.com/in/igorszuniewicz

Kortrijk, Belgium

Applied for

Sound Design MA
FH JOANNEUM
& Kunstuniversität Graz
Institute of Design
& Communication

Document

Selected works
2023 to 2026

ONLINE PORTFOLIO

ACCESSIBLE AT

igorszuniewicz.com

Available from the application deadline until the end of October 2026. Full project pages with embedded videos, audio players, and extended case studies.

ABOUT THIS DOCUMENT

This is my selected sound design and audio programming work from 2023 to 2026. Most of it comes from my Bachelor's in Game Sound Integration at Howest DAE in Belgium. The rest is from my current job as Technical Sound Designer at XR Labs Belgium. Game audio, film scoring, spatial audio, a plugin I wrote, a beatbox game I rebuilt twice, and a thesis on whether AI can generate game sounds fast enough.

Each project page says what the project was, what I did, and what I learned doing it. When the work was collaborative, I name who else was involved and what my specific piece was. Videos, playable builds, and longer case studies are on the online portfolio.

The last three pages are screenshots of every page on the online portfolio, as the FH JOANNEUM application guidelines require.

Igor Szuniewicz

SOUND DESIGNER / AUDIO PROGRAMMER

Legend

SOLO I did everything myself

TEAM Team project, my piece noted

THESIS Bachelor research

JOB Professional work

Total

13

Works listed

Period

2023 to 2026

LIST OF WORKS: AUTHORSHIP AND ATTRIBUTION

Everything listed here is my own work. For team and collaborative projects, I spell out my specific responsibilities on each dedicated project page.

PROJECT	MY ROLE / RESPONSIBILITY	TYPE	YEAR
Shadow Frames	Sole audio designer on DAE + RITCS + Ghent Conservatory team. 200+ Wwise events, C++ footstep and enemy proximity systems, 49-page Sound Bible, full mix	TEAM	2025
Not Today, Darling!	Sole audio designer on 5-person DAE team. MetaSounds engine driven by layered sample variations, mix for 4-player split-screen, 70+ events, team voice recording	TEAM	2025
Transientica	Solo. Started as school Python prototype, rebuilt in Unity C# with KNN classifier written from scratch. FFT, feature extraction, scoring, game loop	SOLO	2025 onward
Ray Animation	Composer and sound designer on DAE graduation animation. Original composer quit 8 days before cinema screening. I wrote full disco-funk score, all SFX, mix	TEAM	2024
AudioQ	Solo. Started as school brief, grew into a full sampler plugin. C++17, JUCE 8, 18 parameters, 10 DSP modules, VST3 + standalone, open source	SOLO	2024 to 2025
Amorak	Solo. Horror animation post-production. 4-animal creature voice (lion, elephant, tiger, hippo), 15+ variations, binaural 3D mix in Dear VR Pro	SOLO	2024
Akantilado	Sound designer on DAE 3D animation. Composer Noah Bots (Ghent Conservatory) wrote the score, I built 200+ custom samples, 5 ambience layers, stereo + 5.1 mix	TEAM	2024
XR Labs Belgium	Technical Sound Designer, sole audio person. Spatial audio pipeline for VR/AR training used by Belgian Defence, Federal Police, Customs	JOB	2025 onward
Richter	Solo. Earthquake short film. Zero commercial libraries. All SFX from Zoom H6 field recording or synthesis (Vital, ReaSynth, TAL Noisemaker)	SOLO	2024
AI Audio Research	Solo bachelor thesis. 68 pages. UE5 + Python backend, 5 AI models tested, 600 generations, 2500 stress tests. Cache-based latency solution	THESIS	2025 to 2026
Sound Design Reel	Solo. 4 redesigns with no commercial libraries: God of War combat, Baldur's Gate 3 spells, P2000 pistol + sci-fi weapon, Fractured Horizon	SOLO	2023 to 2024
Wwise Projects	Solo. 3 middleware studies at DAE: voice limits (zombie shooter), zone transition crossfades (platformer), 2GB Android memory budget (FPS)	SOLO	2024 to 2025
Pause & Deserve	Solo. First UE5 project, 8 weeks. Code, audio, level design, MetaSounds SFX, HRTF spatialization	SOLO	2023

Shadow Frames

HORROR / UE5

Role

Solo Audio Designer
4 months, 2025

Team

DAE game devs, a writer from RITCS (master's thesis), a composer from the Royal Conservatory of Ghent (master's thesis)

Numbers

49 page Sound Bible

32 voice limit

175MB from 1.4GB raw

4 floor surfaces

Tools

Wwise

Unreal Engine 5

C++

Reaper

Shadow Frames

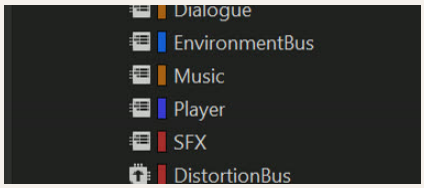
HORROR · UE5 · WWISE · HRTF · 2025 · DAE + RITCS + GHENT CONSERVATORY TEAM

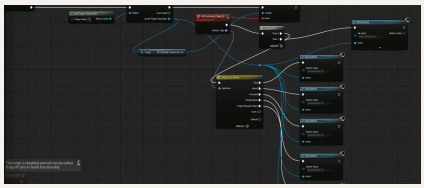


An old man with dementia walks through an abandoned 1960s Irish pub. Everything is black and white. Anomalies keep appearing. Levitating glasses, spinning chairs, drifting shadows. Each one is a fragment of memory. You find them by listening. No HUD, no markers, nothing on screen tells you where to go. Mute the game and you are blind.

Multi-school project. Game devs from Howest DAE, a writer from RITCS working on her master's, a composer from the Royal Conservatory of Ghent also finishing her master's. I was the only audio person. 4 months. I wrote a 49-page Sound Bible so anyone joining mid-project could catch up, then wired up the music triggers so the composer's stems react to proximity and enemy state.

The proximity system was the hardest part. Something always stalks you in the pub. When the enemy hovers near the trigger boundary, early builds toggled the stalker audio on and off every frame. A horrible machine-gun stutter instead of tension. I added a hysteresis state machine with dual thresholds. Start at 70%, stop at 60%. That 10% gap means it cannot flicker. The other big systems: footstep detection that line-traces the floor and reads the physical material (designers can add new surfaces without asking me), three audio stages per anomaly (alert burst, directional hint, sustained loop), and a camera filter that drops everything through a 3kHz low-pass when you raise the Polaroid so breathing gets loud.





WHAT I LEARNED

A 10% hysteresis gap saved the stalker system. The 49-page Sound Bible was mostly for me, honestly. Writing it forced me to actually decide what each sound was doing before I made it.

Wwise · Unreal Engine 5 · C++ · Reaper · HRTF Binaural

Not Today, Darling!

RACING / UE5

Role

Sole Audio Designer
5-person team, 11 weeks, 2025

Numbers

70+ audio events

32 voice limit

450 source files

4 split-screen players

Tools

MetaSounds

Unreal Engine 5

Blueprints

Reaper

Not Today, Darling!

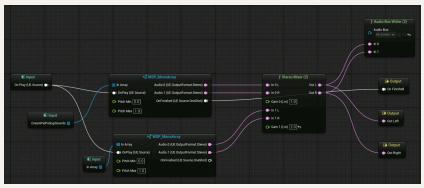
RACING · SPLIT-SCREEN · METASOUNDS · 2025 · 5-PERSON TEAM

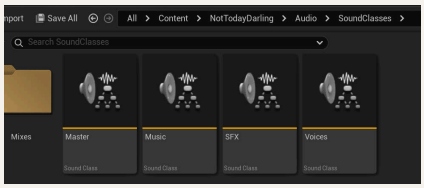


A 4-player side-scrolling racer where grannies smash into each other on go-karts. We had 11 weeks and total creative freedom. Everyone pitched ideas on a whiteboard, we circled what excited us, and built from there. I was the only audio person on a 5-person team, handling the engine sound, the mix, and the granny voices.

This was my first time working with animators, developers, and artists on something real. There were arguments. Conflicts of interest. Before this, nobody at DAE had ever questioned my work. Suddenly everything was up for debate. It sucked at first, but I got it. Collaboration is arguing and compromising without missing the deadline, because four other people are waiting on your stuff.

The engine sound was my first serious MetaSounds project. RPM, throttle, and load modulate layered sample variations. Pitch shifts, volume crossfades between low-revving and high-revving takes, small random timing offsets. Not procedural synthesis, just dynamic sample playback built from recordings. Close your eyes and you know when the kart is accelerating, braking, or sliding. The mix for 4 players was the other hard part. Without structure, 4 engines plus power-ups plus collisions plus granny screams was noise. I built a Sound Class hierarchy where your own kart sits on top, distant karts get attenuated, and power-ups run through concurrency limits. After the fix, nobody mentioned audio anymore. They argued about who rammed who.





WHAT I LEARNED

Collaboration. I shipped with people who disagreed with me and I did not always get my way. That was new. Also: when audio stops being mentioned, it is working. This project is the reason I was ready for Shadow Frames.

MetaSounds | Unreal Engine 5 | Blueprints | Reaper

Transientica

RHYTHM GAME / UNITY

Role

Solo
2025 onward. Rebuilt from school prototype.

Evolution

v1 Python + OSC, SVM

v2 Unity C#, KNN

Technical

KNN classifier, K=5

FFT 512, Hann, radix-2

7 spectral features

Zero allocations on audio path

Transientica

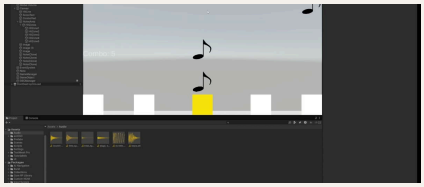
UNITY 6 · C# · KNN · FFT · 2025 ONWARD · SOLO



Beatbox into a mic, play a rhythm game. No MIDI controller, no drum pad, just your mouth. Notes scroll toward you, you make the right sound at the right time, the game judges your timing.

It started as a school research project in Python. Beatbox into a mic, classify the sound with SVM, send results to Unity over OSC. It worked, barely. 180ms latency at first, got it down to 45ms after weeks of optimization. Two processes talking over OSC on localhost. If Python crashed, the game went deaf. Shipping a standalone build meant shipping Python alongside Unity. The prototype was fragile.

So I rebuilt it solo from scratch. Threw out Python, OSC, scikit-learn. Wrote the FFT, feature extraction, and classifier in plain C#. No external ML frameworks, no dependencies I do not control. Switched from SVM to KNN because it trains instantly. Now the classifier trains on your voice specifically. 10 samples per sound, takes a few seconds, and it learns how you do a kick versus how I do a kick. Perfect ±60ms, Great ±130ms, Good ±200ms. All timing runs on AudioSettings.dspTime, not frame time.





WHAT I LEARNED

The prototype worked but did not feel like a game. Python and OSC were fine for proof of concept and wrong for shipping, and I should have noticed that earlier. I spent more time than I want to admit porting the FFT to C# before trusting it.

Unity 6 · C# · URP · Python (v1) · scikit-learn (v1)

Ray Animation

GRADUATION FILM SCORE

Role

Composer & Sound Designer
8 days, 2024

Deliverable

Full disco-funk score

All sound design

Final mix

Tools

Pro Tools

Logic Pro

Omnisphere

Waves J37 tape

Ray: Graduation Animation Score

ANIMATION DISCO-FUNK RESCUE JOB HOWEST DAE 2024 · DAE GRADUATION TEAM

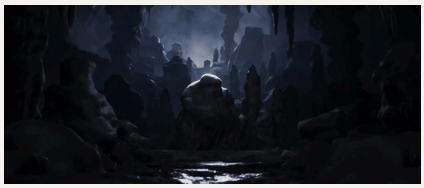


A stone giant wakes up in a cave and discovers dance. That is the film. The original composer dropped out 8 days before the cinema screening. No material, no stems, no sketches. Just gone. I got the call and said yes.

8 days. Full disco-funk score from scratch, all the sound design, final mix. I pulled the animatics into Pro Tools, placed sync markers by hand, and locked every musical hit to Ray's movements frame by frame. The tempo follows Ray's dancing, not a click track. When he stomps, the kick lands. When he grooves, the bass follows. Chic, Earth Wind & Fire references. Slap bass, brass, sax, DJ scratches between sections, whole mix through Waves J37 tape saturation for warmth.

Day 1 and 2: theme sketches and sync markers. Day 3 to 5: full arrangement, all instruments tracked. Day 6 and 7: sound design and foley. Day 8: final mix and cinema delivery. Three rounds of feedback with the animation team, tightening the timing each pass. Screened at the DAE cinema.





WHAT I LEARNED

Writing to deadline is not something 12 years of classical piano prepared me for. Three feedback passes, first drafts had to hold. I still do not know if the score is actually good or just tight; I only know it landed at the cinema and the animators were happy.

Pro Tools | Logic Pro | Omnisphere | Waves J37

AudioQ

VST3 / STANDALONE

Role

Solo
2024 to 2025. Started as school brief.

Numbers

18 parameters

10 DSP modules

VST3 + Standalone

Stack

C++17

JUCE 8

StateVariableTPT

APVTS

Source

github.com/Szunias/AudioQ · MIT

AudioQ: Granular Sampler Plugin

C++17 JUCE 8 VST3 DSP 2024 TO 2025 · SOLO



I kept loading a granular plugin, then a filter, then a tremolo, routing audio between all three just to mangle one sample. Annoying. So I wrote my own plugin that does all of it in one window. Drag in any WAV, AIFF, or MP3, twist it with granular, filters, compression, bitcrushing, stutter, reverse, and automate every knob from your DAW.

It started as a school brief. The first version had granular, tremolo, and filters. Then I kept adding because I kept needing things. Compressor (because granular has wild volume spikes). Bitcrusher (because I wanted lo-fi crunch). Stutter and reverse from a shared 1-second history buffer. Dry/wet mix. Region-based looping on the waveform. A real-time waveform display next to the static one. 18 automatable parameters total. The filters use JUCE's StateVariableTPT topology because naive direct-form filters warp and blow up under fast modulation.

C++17, JUCE 8, compiles to VST3 and standalone. Volume safety monitor that auto-stops playback if peaks exceed +12dB, because granular mode blew my ears out too many times while testing. Source on GitHub, MIT license.





WHAT I LEARNED

Writing DSP from scratch was harder than reading about it and easier than I thought it would be once I started. StateVariableTPT gave me the first intuition for why naive filters blow up under modulation. I am still not sure the compressor is actually good, just that it does what I need.

C++17 JUCE 8 CMake Visual Studio

Amorak

HORROR ANIMATION / 3D

Role

Sound Designer
4 weeks solo, 2024

Creature

Lion (chest-rattle low)
Elephant (mass)
Tiger (snarl)
Hippo (rasp)

Tools

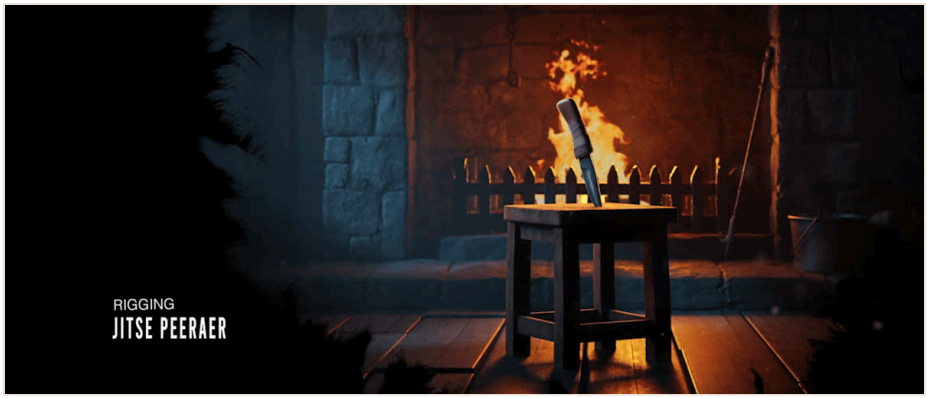
Reaper
Dear VR Pro
Krotos Studio
IEM Suite
Logic Pro

Mix

Binaural 3D
Headphones required

Amorak: Horror Animation

3D ANIMATION · CREATURE · BINAURAL · 90 SECONDS · 2024 · SOLO

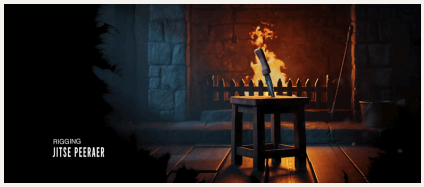


A man hears something outside. Hides in the basement. Traps the creature. 90 seconds of horror, and I did every sound you hear. Put headphones on. The creature sounds like it is right behind you. That was the whole point.

The creature voice is four animals layered together. Lion for the chest-rattling low end, elephant for mass, tiger for the snarl, hippo for that raspy wheeze. Each layer sits in its own frequency range so they do not fight each other. I pitched and time-stretched each source separately in Krotos Studio, then bounced 15+ vocal variations so no two growls sound the same. The creature never loops.

I mixed everything in binaural 3D with Dear VR Pro. On headphones, the creature does not just get louder, it gets closer. Water drips from above left, the creature moves behind you, reverb tail shortens as the camera pushes in. One rule for the mix: the creature always wins, but the man's breathing stays audible. Sidechain ducking so breathing never gets buried.





WHAT I LEARNED

Mixing for headphones is different than for speakers, and I mostly figured that out by making mistakes the first time. The 3D position did more heavy lifting than I expected, so I kept pulling volumes back. The creature vocal took longer than anything else in the project, including two days on the layering that I am still not fully happy with.

Reaper · Dear VR Pro · Krotos Studio · IEM Suite · Logic Pro

Akantilado

3D ANIMATION / 5.1 MIX

Role

Sound Designer
4 months, 2024, Howest DAE

Composer

Noah Bots
Ghent Conservatory

Numbers

200+ custom samples

5 ambience layers

Stereo + 5.1 surround

Tools

Reaper

Foley

5.1 Surround

Akantilado: Jungle Animation

3D ANIMATION · 5.1 SURROUND · FOLEY · HOWEST DAE · 2024 · DAE
+ GHENT CONSERVATORY TEAM



Two cats chase a golden orb through a jungle canopy. One of them goes over the cliff. I built every sound in this film from scratch. 200+ custom samples. Paw steps on wet leaves, claws digging into bark, plants whipping back after something brushes past them. Noah Bots from Ghent Conservatory composed the score, I did everything else.

The jungle itself is five ambience layers running at all times. Leaf rustle, distant animal calls, running water, wind, and a low room tone that gives the space depth even in silence. They crossfade based on camera distance. Close to the ground, you hear insects and leaf detail. Pull up to the canopy, it is wind and distant bird calls. There is always something going on in the background.

The two cats needed to sound like different animals, not the same sample pitched up and down. Orange gets a lower, throatier voice and heavy footfalls. You feel him land. Pink is higher, quicker, lighter on every surface. You can tell which cat is moving before you see them. I delivered stereo first for the rough cut, then a 5.1 surround mix for cinema. Three spatial zones: canopy above, ground level around you, distance behind. When the camera is in the trees, the surround channels fill with leaves. When it drops to the ground, the surrounds fill with detail.





WHAT I LEARNED

The jungle had to sound different from the canopy than from the ground, and I did not really plan for that, I just kept noticing it sounded wrong and rebuilding the ambience layers. First time mixing for real cinema 5.1. The downmix from surround back to stereo was messier than I thought it would be.

Reaper · Foley · 5.1 Surround · Custom recording

XR Labs Belgium

VR/AR SPATIAL AUDIO

Role

Technical Sound Designer
Sole audio, 2025 onward

End Users

Belgian Defence

Federal Police

Customs

Pipeline

Unity

Steam Audio

Perforce

XR Labs Belgium: VR/AR Training Sims

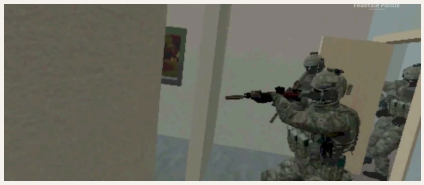
VR / AR · UNITY · STEAM AUDIO · SPATIAL · 2025 ONWARD · CURRENT PROFESSIONAL ROLE



XR Labs is a Belgian initiative where defence, federal police, and customs build VR/AR training simulations together. I am the sole audio person. I build and maintain all the spatial audio systems across their projects.

Trainees need to locate threats by sound alone. A gunshot has to come from the right direction, at the right distance, through the right walls. If the spatial audio is off by even a few degrees, someone trains wrong. I set up Steam Audio propagation so sounds reflect and occlude based on actual room geometry, not baked approximations. I work with Unity and Steam Audio, version-controlled through Perforce.

The specifics are confidential, but the work is mostly spatial accuracy problems. Making sure a voice behind a closed door sounds behind a closed door, not next to you. Tuning occlusion curves iteratively because VR proximity amplifies every mismatch. When trainees stop thinking about the audio and just react, that is when the system is working.





WHAT I LEARNED

Game players forgive wrong spatial audio because the stakes are low. Trainees do not. They train wrong and carry that into real situations. I had to retune what I thought "close enough" meant. Baked reverb is not close enough when someone's job depends on where they hear the voice coming from.

Unity · Steam Audio · Perforce · Reaper

Richter

EARTHQUAKE SHORT FILM

Role

Sound Designer & Composer
2024

Rule

Zero commercial libraries
Every sound recorded or synthesized

Sources

Zoom H6 field recording

Vital synth

ReaSynth

TAL Noisemaker

Richter: Earthquake Film

SHORT FILM · FOLEY · SYNTHESIS · NO LIBRARIES · 2024 · SOLO

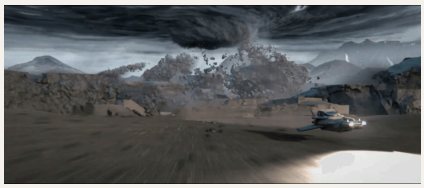


Student short film about an earthquake. The brief came with one constraint: no commercial sample libraries. If a sound appears in this film, I either recorded it or synthesized it from nothing.

That meant walking around campus with a Zoom H6, recording footsteps on wood, concrete, and carpet. Slamming doors. Crinkling paper. Dragging cloth across surfaces. Anything that could become "building falling apart" later in the edit. Ceiling cracks were impossible. I walked campus for days looking for the right one, then stomped on a dry wooden pallet and that was it.

The earthquake itself is synth drones from Vital and ReaSynth pitched way down, with granular-stretched foley on top. Pure low-end rumble sounds fake. Adding physical texture, stretched door creaks, concrete scrapes, makes the shaking feel real. I also wrote a short orchestral cue, strings and brass, timed to the scene cuts. The biggest mix problem: the quake rumble ate every word of dialogue. I carved a pocket around 2 to 4kHz with EQ automation and added sidechain ducking so the rumble dips whenever someone speaks. A full day of Reaper automation to get the balance right.





WHAT I LEARNED

Pure synth rumble never sounded right. Adding stretched foley underneath fixed it, but I spent a full extra day on dialogue ducking before the mix actually worked.

Reaper · Cubase · Zoom H6 · Vital · TAL Noisemaker

AI Audio Research

BACHELOR THESIS

Role

Solo Researcher
Howest DAE, 2025 to 2026

Supervisors

De Meulemeester Roel

Van der Kelen Cedric

Numbers

600 generations

2500 stress tests

5 models tested

68 page thesis

Hardware

RTX 5070 8GB VRAM
Ryzen 9 7940HS, 64GB RAM

Generative AI for Real-Time Game Audio

THESIS · UE5 + PYTHON · AI MODELS · CACHING · 2025 TO 2026 · SOLO RESEARCH

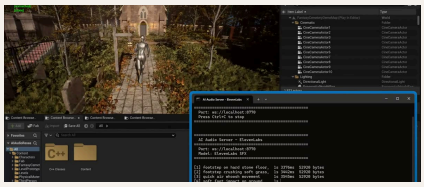


Every game sound is a WAV file someone recorded. A big game ships with thousands of them. Gigabytes of "sword hits metal." What if AI could generate them on the fly instead? No more shipping audio banks the size of a movie. The problem is timing. Games run at 60fps, 16.7ms per frame. If a gunshot arrives 100ms after you pull the trigger, players feel it. I wanted to measure exactly how far off we are.

UE5 talks to a Python backend over WebSocket. I tested five approaches. Procedural DSP (just math, 15 to 25ms, sounds like a ringtone). AudioGen (Meta's text-to-audio, 2.1 to 2.8s). MMAudio Large and Small (video-to-audio models, 1.4 to 3.4s). ElevenLabs cloud (sounds best, 1.8 to 3.5s, latency is a lottery). 24 prompts, 5 categories, 600 generations, plus 2500 stress tests. Every model failed the 150ms target by at least an order of magnitude.

Then I looked at the cache logs. I had added caching early on just to stop regenerating the same sound. Delivery times from cache: 35 to 90ms. That is fast enough. A footstep that takes 2400ms to generate, served from cache, takes 45ms. Predicted before the player moves, under 10ms from the engine buffer. A 95 to 98% reduction. The models were not the answer. The cache was. Stop trying to make AI faster and start asking when to generate. Loading screens, zone transitions, quiet menu moments. Generate the sounds before the player needs them. By the time they move, the sound is already in memory.





WHAT I LEARNED

Six months optimizing model speed, one afternoon reading cache logs, and the bottleneck was somewhere I was not looking. Whether this scales to a competitive multiplayer shooter, I do not know. It works for the prototype and for most single-player pacing, which is where I want to take it next.

Unreal Engine 5 · C++ · Python · PyTorch · WebSocket · CUDA

Sound Design Reel

4 REDESIGNS

Role

Solo Sound Designer
2023 to 2024, 12 months

Redesigns

God of War combat

Baldur's Gate 3 spells

P2000 + sci-fi weapon

Fractured Horizon (first)

Rule

No commercial libraries
Mic or synth only

Sound Design Reel: 4 Redesigns

REDESIGNS FOLEY SYNTHESIS NO LIBRARIES 2023 TO 2024 · SOLO

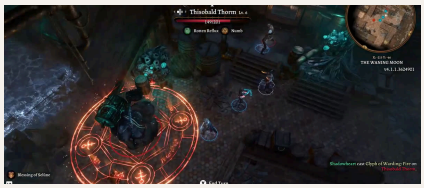


I took game scenes I knew by heart, muted the original audio, and rebuilt them using only my own recordings and processing. Four redesigns over 12 months. The rule across all four: no commercial libraries. Everything starts from a microphone or a synth oscillator.

God of War combat. Axe hits are metal-on-meat. I hit a pork shoulder with a wrench and that became the base layer. Shield blocks have a sharp crack because I compressed the transient hard and let the tail ring out. The enemy screams took longest. First attempts sounded cartoonish. I ended up pitching down my own voice recordings and blending them with animal sounds until it stopped being funny and started sounding genuinely disturbing. **Baldur's Gate 3 spells.** Fire starts with a gas-burner recording pitched down an octave, synth crackle on top for the magical layer. That two-layer approach (physical on the bottom, synthetic on top) became my formula for all the spell sounds. Ice is mostly granular synthesis with a slow attack so it sounds like it is forming.

P2000 pistol + sci-fi weapon. Same exercise, opposite directions. P2000 is foley and field recordings. Every layer recorded in the real world. The sci-fi weapon is pure synthesis and spectral processing, no microphone involved at all. Sounds like electricity arcing through glass. Doing both at once made it clear how different the two approaches are. **Fractured Horizon.** My first sound design project ever. Honestly rough, basic spaceship hums and door sounds. But this is where I learned how much timing matters. A door sound 3 frames late looks completely wrong.





WHAT I LEARNED

3 frames of bad sync ruins everything, and your brain catches it instantly even if you cannot explain why. The no-libraries rule was annoying at first and then it changed how I walk around a kitchen. A gas burner pitched down an octave ended up being my favorite fire sound.

Reaper | Logic Pro | Granular Synthesis | Field Recording | Foley

Wwise Projects

MIDDLEWARE / 3 PROJECTS

Role

Solo Audio Designer
3 school projects, 2024 to 2025

Projects

Boney Boogaloo (zombie FPS)

Ellen (platformer)

Stomach Aches (Android)

Biggest Win

80% memory saved via Wwise Silence plugin

Wwise Projects: 3 Middleware Studies

WWISE UNITY RTPCS OPTIMIZATION 2024 TO 2025 · SOLO



Three solo school projects at DAE between 2024 and 2025. A zombie shooter where I hit the 256 voice limit on day one. A platformer where zone transitions sounded like someone flipping a switch. An Android game that crashed on load because all audio was in memory at once. None of them shipped, but each one broke something I had to actually fix.

Boney Boogaloo (zombie wave shooter). 256 voices in Wwise fills up fast when every gunshot, spawn, and death scream takes a slot. Sounds were cutting out mid-play. Took me an hour to realize the problem was not my code, it was Wwise running out of voices. I gave each sound 6 variations so nothing repeats, then set priorities so nearby gunshots always win over a zombie groaning in the distance. 44 sounds total, and nothing cuts out anymore. **Ellen (platformer).** I focused on making spaces sound real. A waterfall stays loud from far away, footsteps disappear after a few meters. Each sound type has its own distance curve. The hard part was zone transitions. Forest to cave crossfades over 6.5 seconds. Shorter felt rushed, longer felt laggy. The reverb changes with the crossfade too. Big rooms sound big, tight corridors sound tight. No hard cuts.

Stomach Aches (FPS platformer, Android). Set inside a human body, targeting 2GB RAM Android phones. First build crashed on load. All audio loaded at once. I split it into zones so only the current area stays in memory. Brain takes 670KB, Mouth 384KB, Stomach 2.8MB. For quiet moments I replaced real audio files with generated silence, cutting memory use by 80%. The waterfall has 3 distance layers so it sounds different at 5 meters versus 30. Total audio per zone never exceeds 3MB.





WHAT I LEARNED

Voice limits, distance curves, memory budgets. These are not things you plan for on paper, they are things you panic-fix at 2am because the build crashed on your phone during a playtest. I do not think I would have learned any of this without the builds failing first.

Wwise Unity SoundBanks RTPCs ADPCM

Pause & Deserve

HORROR / FIRST UE5

Role

Solo Developer
8 weeks, 2023

The Deal

Pause freezes enemy 5s

Enemy gets 1.2x faster each pause

After 5 pauses, ~2.5x speed

Tools

Unreal Engine 5

MetaSounds

Sound Cues

UE5 HRTF

Pause & Deserve: First UE5 Project

HORROR FIRST UE5 METASOUNDS HRTF 2023 · SOLO (8 WEEKS)



I had never opened Unreal Engine before this. Gave myself 8 weeks to learn it by building something from scratch. Code, audio, level design, everything solo. The result is a horror game built around one bad deal: you can pause to freeze the enemy for 5 seconds, but every time you do, it gets 20% faster. The math catches up quickly, and there is a point where you cannot outrun it anymore.

Audio carries the whole experience. There are barely any visuals telling you what to do. I built the atmosphere entirely from foley and field recordings, handled spatial audio through UE5's HRTF plugin, and used MetaSounds for the SFX with Sound Cues for triggers. Layers of rustling grass, cracking branches, wind through trees. Random paranoia sounds mixed in. A twig snap behind you that might be nothing. You can never tell what is the forest and what is the enemy.

With headphones on, UE5 HRTF puts footsteps behind you, wind passing from left to right, and you actually check over your shoulder when a branch cracks back there. Players started rationing pauses like ammo.





WHAT I LEARNED

I spent days synthesizing creepy forest sounds in MetaSounds. Then I recorded boots on dirt, hands on bark, fabric rustling, and was done in five minutes. I should have started with the microphone.

Unreal Engine 5 MetaSounds Sound Cues UE5 HRTF Blueprints

Online Portfolio

WEBSITE DOCUMENTATION

URL

igorszuniewicz.com

Access

Available from application deadline until end of October 2026

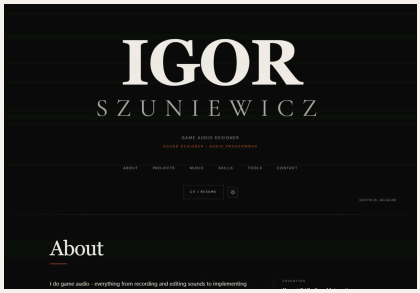
Note

Screenshots per FH JOANNEUM requirements. Videos and interactive content captured as static frames on page 19.

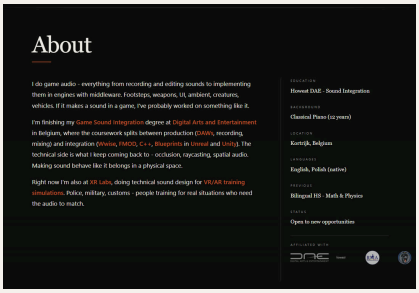
ONLINE PORTFOLIO SCREENSHOTS (1 OF 3)

Screenshots of all main navigation pages of igorszuniewicz.com, as required by the FH JOANNEUM application guidelines.

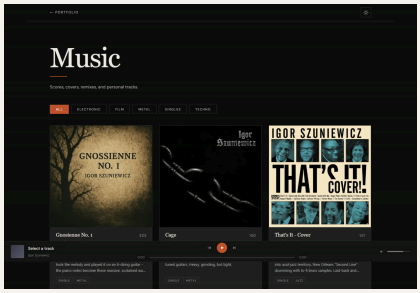
Tip: zoom in (Ctrl + Scroll) or right-click any image and choose "View Image" / "Open image in new tab" to read the screen content at full resolution.



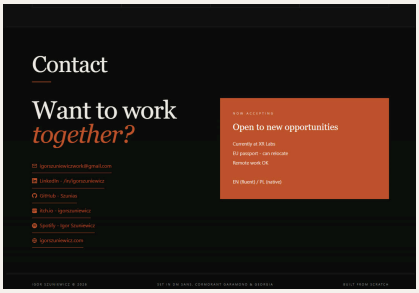
Home
igorszuniewicz.com



About
igorszuniewicz.com/about



Music
igorszuniewicz.com/music



Contact
igorszuniewicz.com/contact

Online Portfolio

PROJECT PAGES

Pages

Shadow Frames

Not Today, Darling!

Transientica

Ray Animation

AudioQ

Amorak

Akantilado

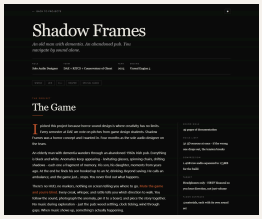
Richter

Sound Design Reel

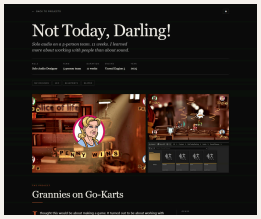
ONLINE PORTFOLIO SCREENSHOTS (2 OF 3)

Individual project pages on igorszuniewicz.com.

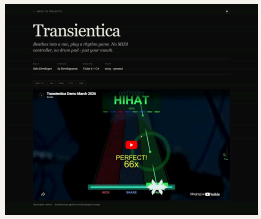
Tip: zoom in (Ctrl + Scroll) or right-click any image and choose "View Image" / "Open image in new tab" to read the screen content at full resolution.



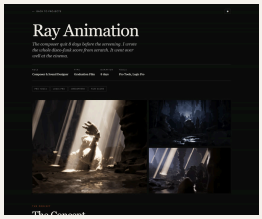
Shadow Frames
/shadowframes



Not Today
/nottoday



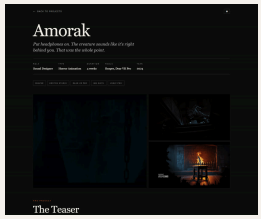
Transientica
/transientica



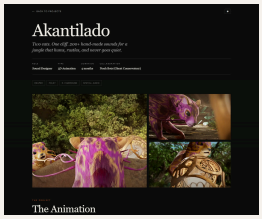
Ray Animation
/ray



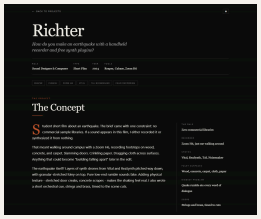
AudioQ
/audioq



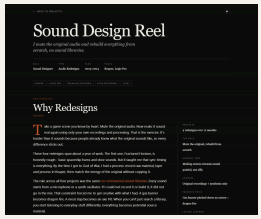
Amorak
/amorak



Akantilado
/akantilado



Richter
/richter



Sound Design
/sounddesign

Online Portfolio

REMAINING PAGES + VIDEO

Pages

XR Labs

Wwise Projects

Pause & Deserve

AI Audio Research

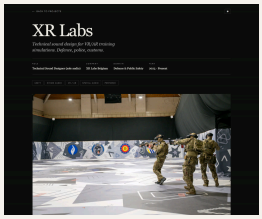
Video Content

Videos on the online portfolio shown here as static frame captures, as FH JOANNEUM requires.

ONLINE PORTFOLIO SCREENSHOTS (3 OF 3)

Remaining project pages and static captures of embedded video content.

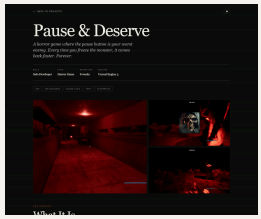
Tip: zoom in (Ctrl + Scroll) or right-click any image and choose "View Image" / "Open image in new tab" to read the screen content at full resolution.



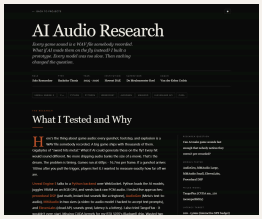
XR Labs
/xrlabs



Wwise Projects
/wwiseprojects



Pause & Deserve
/pausedeserve



AI Audio Research
/ai-audio-research



Shadow Frames · Trailer
embedded video frame



Amorak · Creature
embedded video frame



Ray · Animation
embedded video frame



AudioQ · Demo
embedded video frame



Not Today · Trailer
embedded video frame

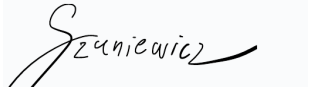
END OF DOCUMENT 2026

THANK YOU FOR YOUR TIME

Thank you.

Contact
igorszuniewiczwork@gmail.com
+48 605 117 516
linkedin.com/in/igorszuniewicz

Online Portfolio
igorszuniewicz.com



→ igorszuniewicz.com

Kortrijk, 2026